

CS23334 Fundamentals of Data Science Lab

Experiment 3.a

Conduct an experiment to show data visualization using line plot

Description: Take any sample data either through csv file or data fetched directly through code.

Code:

```
import matplotlib.pyplot as cricket

Overs=list(range(5,51,5))

Indian_Score=[30,55,90,129,165,200,239,270,310,350]

Srilankan_Score=[25,70,90,120,140,170,195,220,255,279]

cricket.plot(Overs,Indian_Score)

cricket.plot(Overs,Srilankan_Score)

cricket.show()

cricket.title("INDIA Vs SRILANKA")

cricket.xlabel("Overs")

cricket.ylabel("Score")

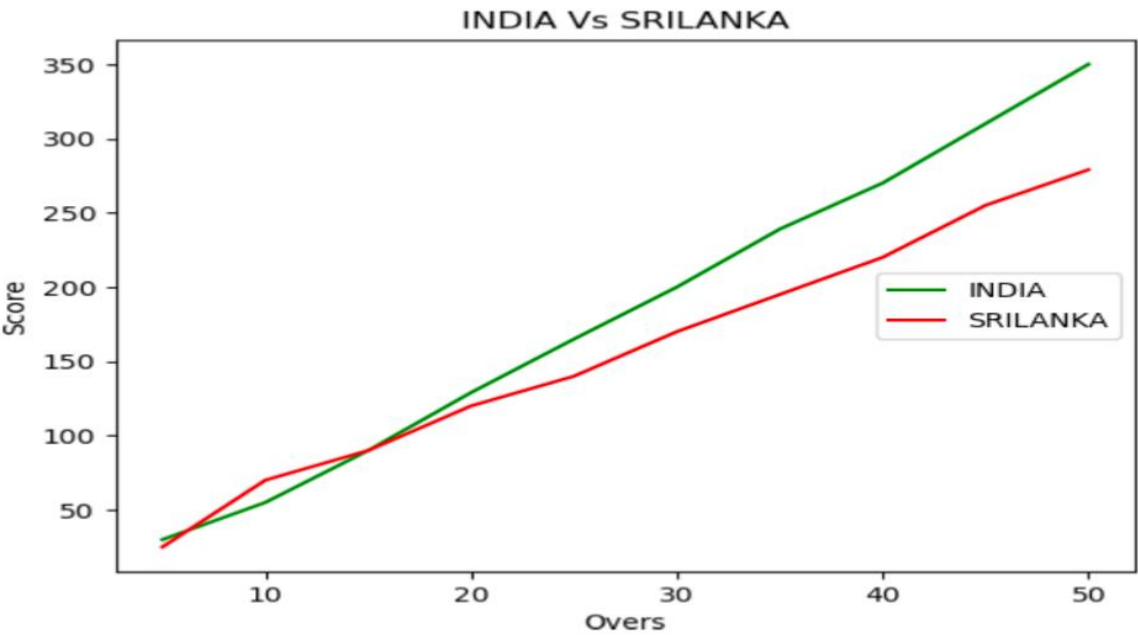
cricket.legend()

cricket.plot(Overs,Indian_Score,color="green",label="INDIA")

cricket.plot(Overs,Srilankan_Score,color="red",label="SRILANKA")

cricket.legend(loc="center right")
```

OUTPUT:



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Experiment 3.b

Conduct an experiment to show data visualization using bar chart

Description: Take any sample data either through csv file or data fetched directly through code.

Code:

```
import matplotlib.pyplot as hscmark

import numpy as np

Names = ['SHREE', 'DEV', 'KEERTHI', 'PRIYA', 'SHAN', 'KUMARAN']

xaxis = np.arange(len(Names))

Percentage_hsc = [96, 91, 94, 75, 45, 81]

hscmark.bar(Names, Percentage_hsc)

hscmark.xticks(xaxis, Names, rotation=45)

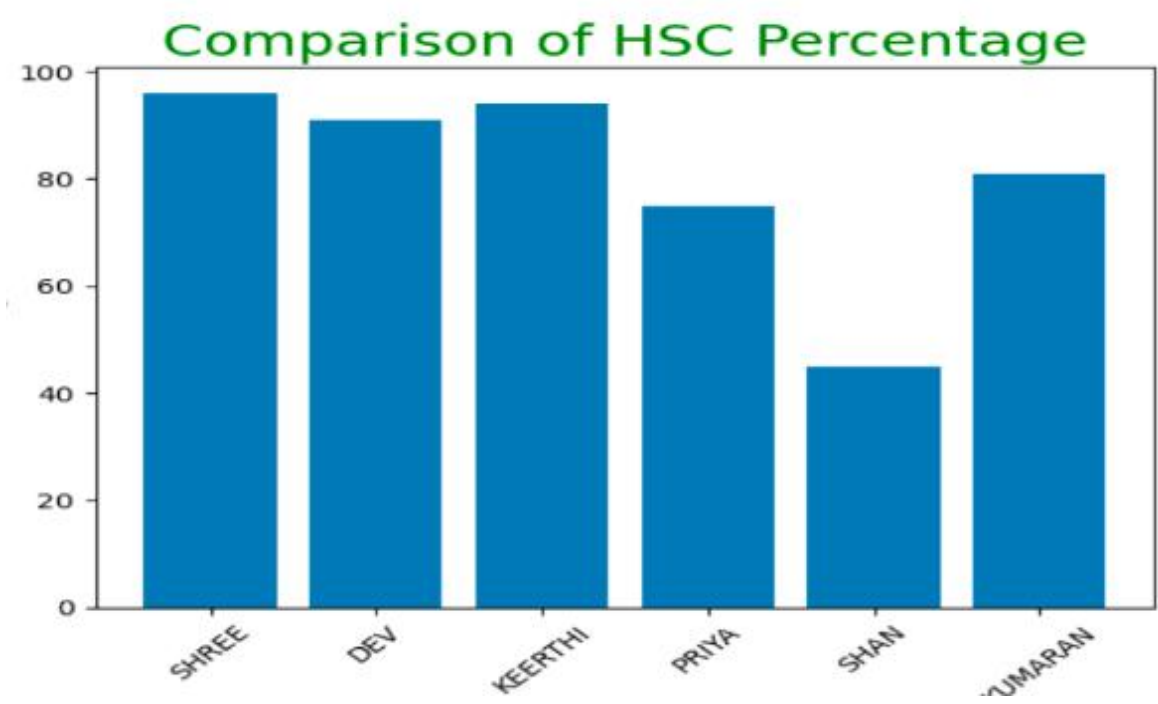
hscmark.xlabel("Names of Pupil")

hscmark.ylabel("Percentage")

hscmark.title("Comparison of HSC Percentage", fontsize=20, color="green")

hscmark.show()
```

OUTPUT:



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Experiment 3.c

Conduct an experiment to show data visualization using pie chart

Description: Take any sample data either through csv file or data fetched directly through code.

Code:

```
import matplotlib.pyplot as election

labels = ['CANDIDATE 1', 'CANDIDATE 2', 'CANDIDATE 3', 'CANDIDATE 4']

Votes = [315, 130, 245, 210]

colors = ['green', 'yellow', 'red', 'orange']

explode = (0.2, 0, 0, 0)

election.pie(Votes, labels=labels, colors=colors, explode=explode, autopct='%0.2f%%')

election.title('Election Results')

election.show()
```

OUTPUT:

